MILITARY SPECIFICATION

TUBE, CONNECTING, SURGICAL APPARATUS, GLASS

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a straight, glass connecting tube, suitable for use with ¼ inch rubber tubing as used on irrigating apparatus.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extend specified herein:

SPECIFICATIONS

FEDERAL

PPP–B–566 — Boxes, Folding, Paperboard.
PPP–B–621 — Boxes, Wood, Nailed and Lock Corner.
PPP–B–636 — Box, Fiberboard.
PPP–B–676 — Boxes, Set-Up, Paperboard.

MILITARY

MIL–L–10547 — Liners, Case, Waterproof.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 626 Glassware; Methods for Sampling and Testing.

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129 — Marking for Shipment and Storage.

(Copies of specifications and standards, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The connecting tubes shall be fabricated of good quality clear borosilicate glass properly annealed and of such composition as to meet the requirements hereinafter specified.

3.2 Dimensions and construction. Dimensions and construction shall be as shown in figure 1. The ends shall be square cut and smoothly fire-polished.

3.3 Thermal shock resistance. Sample tubes, selected at random in accordance with 4.2.1 and 4.2.3, shall be subjected to the thermal shock test specified in 4.4.1 and shall not crack, chip or break as a result of such testing.

3.4 Workmanship. Workmanship shall be first class throughout. Glass shall be smoothly and symmetrically finished. Glass shall be uni-

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formly distributed, properly annealed, and free from seeds, blisters, and other defects or irregularities which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Inspection.* Inspection as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.* The inspection lot shall be determined by the cognizant government inspector and shall be based on the unit of product being one tube (end item).

4.2.2 *Sampling for examination.* A random sample of tubes shall be selected from each lot offered to the Government in accordance with Standard MIL-STD-105. Examination for dimensions shall be in accordance with Standard MIL-STD-105 and as indicated in table I.

4.2.3 *For tests.* Sampling for tests shall be conducted in accordance with Standard MIL-STD-105 and as indicated in table I.

4.3 Examination. The tubes shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Thermal shock test.* The shock test shall be performed in accordance with method 401 of Federal Test Method Standard No. 626 with oven maintained at 100°C., except that samples selected for test shall not be divided into two equal lots as indicated in paragraph 2 (Procedure) of method 401. In lieu thereof, one sample only shall be selected in accordance with 4.2.3, above, and lot disposition based thereon.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit of issue.* Twelve (12) tubes, as specified, constitute one unit of issue.

5.1.2 *Packaging quantities.* The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

5.1.2.1 *Packaging variation permitted.* If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

TABLE I. *Sampling Criteria*

Component	Characteristic	Req.	Test procedure	Unit of product	Insp. level	AQL (percent defective)
Tube	Thermal shock test	3.3	4.4.1	1-tube	I3	2.5
	Dimensions	3.2	..	1-tube	I3	4.0

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TABLE II. *Packaging quantities*

Unit package	Intermediate package	Exterior container
1 unit	12 units	288 units

5.1.3 *Level A.*

5.1.3.1 *Unit package.* Each unit shall be packaged in a carton of appropriate size constructed in accordance with Specification PPP-B-566 or PPP-B-676. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or prevent damage to the contents. Each carton shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.* Intermediate package shall be a carton of appropriate size constructed in accordance with Specification PPP-B-636 or PPP-B-676. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.* Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 **Packing.**

5.2.1 *Level A.*

5.2.1.1 *Exterior container.* Exterior container shall be constructed in accordance with Specification PPP-B-621, table IV.

5.2.1.2 *Case liner.* Each level A box shall be lined with a waterproof case liner conforming to Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* Strapping, when required, shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.* Exterior container shall be constructed in accordance with Specification PPP-B-621, table II or PPP-B-636, type I, class 1, fabricated from not less than 275-pound test fiberboard.

5.2.3 *Level C.* The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Unit package.* Each unit package shall be marked with stock number, item identification, unit, quantity, and name of contractor.

5.3.2 *Intermediate package.* Each intermediate package shall be marked in accordance with Standard MIL-STD-129.

5.3.3 *Exterior container.* Exterior container shall be marked in accordance with Standard MIL-STD-129.

6. **NOTES**

6.1 **Ordering data.** Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

6.3 **Transportation description.** Transportation description applicable to this item is:

Glassware, Chemical Laboratory, in boxes.

Carload minimum weight 18,000 pounds subject to Rule 34 of the Uniform Freight Classification.

Truckload minimum weight 18,000 pounds per National Motor Freight Classification.

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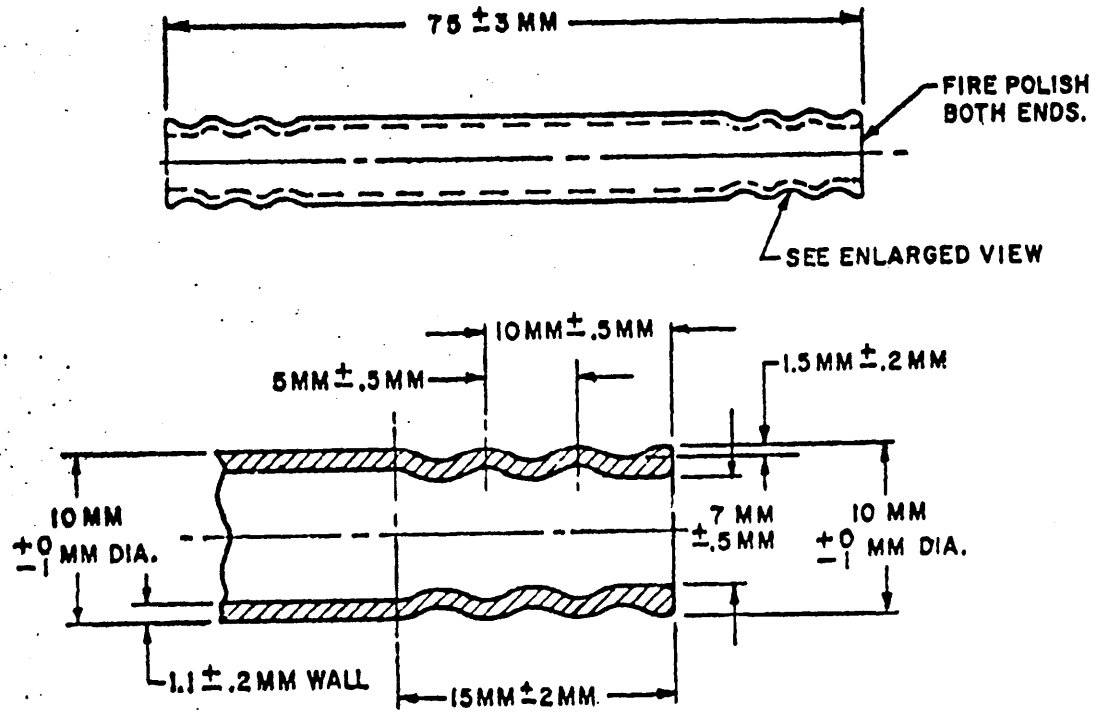


FIGURE I.

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